

CSC 480: Research, Writing and Presentation tips

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General Research Tips

How to choose a topic for my talk / project?

- Familiarize yourself with the main conferences and journals in your area of interest:
- Examples of general AI/ML conferences: AAAI, NeurIPS, IJCAI, ICML
- Evolutionary computation / bio-inspired: Gecco, Evostar, ALIFE
- Games: CoG, AIIDE, FDG
- Computer Vision: CVPR
- Computational Creativity / Art / Music: ICCA (International Conference on Computational Creativity), EvoMUSART, AIMC

How to choose a topic for my talk / project?

- Conferences often have competitions and tutorials. These tend to be more accessible and will often include sample code
- Workshops at conferences tend to be more focused than the general proceedings and also include work that is still ongoing (so there might be more opportunity to improve on it)
- Examples:
 - CoG competitions
 - AAAI tutorials
 - Procedural Content Generation (PCG) workshop at FDG
- Note: if you base your work on a competition, it does **not** matter to me whether the competition is still ongoing or if you can make its deadline, only that the supporting resources exist.

How to choose a topic for my talk / project?

- A good topic (for the project) should be:
 - Accessible: ideally, there's some existing implementation or similar resources you can use as a starting point
 - Reasonably complex and novel
 - Example: merely reproducing a blog post would not be novel enough, but applying an existing algorithm with a few tweaks, or in an untried context would
 - Achievable with incremental goals (avoid catastrophic failure)

Google Scholar is your friend!

- Search by author / year / relevance etc
- Often includes open access pdf versions of the papers
- Handy citation tool (various standards, as well of BibTex)
 - Speaking of BibTex...

LaTeX (and Overleaf) is your friend

- Your final report will be formatted as a conference paper
- LaTeX is typically the easiest way to do it.
- BibTex also helps you manage your references easier
- Lots of conferences publish LaTeX templates that are a good learning tool
- Overleaf makes it easier to edit your document online and to collaborate with others

Maintain a reading log!

- Create a spreadsheet to track your paper readings
- Record name, author, year, conference and link to every paper you read
- Include one or more columns with your personal observations
 - Example: a summary, a list of algorithms, connections to your research topic, connections to other papers...
 - It's ok to keep track of papers you've only skimmed, or that you want to read in the future too!
- Follow through references that are inference (or at least record them to read later)

Writing tips (proposal & report)

How much detail to include?

- Do not assume high levels of familiarity with your methods
 - Reader is technically competent, but not an expert in your problem
- Focus on:
 - inputs/outputs
 - main parameters/hyperparameters
 - main components / steps
 - relevant data structures (the structures themselves don't need to be detailed)
 - how to train the system
 - how to interact with the system

Avoid clutter!

“The secret of good writing is to strip every sentence to its cleanest components. Every word that serves no function, every long word that could be a short word, every adverb that carries the same meaning that’s already in the verb, every passive construction that leaves the reader unsure of who is doing what—these are the thousand and one adulterants that weaken the strength of a sentence. And they usually occur in proportion to the education and rank.”

William Zinsser in *On Writing Well*, 1976

Avoid clutter!

- Adverbs can usually be cut
 - “very inefficient” -> “inefficient”
 - “conclusively demonstrate” -> “demonstrate”, “show” (if not conclusive, “suggest”, “indicate”)
- The passive voice should be avoided! 🙄
 - Leaves the reader wondering who did what
- Prefer simpler words/expressions to longer ones
 - “study the main principles of X” -> “study X”
 - “provide a review of X” -> “review X”
 - “Utilize” -> “use”
- This can be done in a “final pass” over the text: aim to cut total length by ~10%

References and labels

- Every reference should be numbered and referenced in the text.
 - Google scholar is your friend for academic papers
 - For non-academic papers, make sure you have author name, title, publication date, access date
- Every table/image should also be numbered and referenced in the text
 - “Table 2” rather than “The table below”
 - Tables/images also need have a caption that makes it clear what they are about if the reader forgot where it was referenced in the text
 - Use the text and/or caption to interpret anything that may be ambiguous: is a bigger or smaller value better? In a match-up table, does the number refer to the row agent or column agent’s win rate?

Jargon and ambiguity

- Avoid jargon if possible, especially if ambiguous
- For example, “data” could mean:
 - Your agent’s observation at a point in time
 - Your agent’s observation history
 - Your entire dataset
 - A point in your dataset (beware singular vs plural “data”)
 - Your training/validation/test dataset
 - Your experiment’s results (“our data shows...”)

Jargon and ambiguity

- Don't use two different words to denote the same concept
 - Variable
 - Feature
 - Attribute
 - Property...
- Especially if you're referencing another source!

Avoid double negatives!

- They're a big no-no

Presentation tips

Telling a good story

- Your presentation will tell a story, and it should consist of three phases
 - **The opening**
 - **The body**
 - **The closing**
- Your main message should echo through all three phases

The opening (2~4 slides)

- State your main message (“we developed an agent to play tic-tac-toe...”)
- Provide the appropriate context
- Answer the “why” question
 - Why is your problem interesting and/or challenging?
 - How does it relate to class objectives?
 - What where you hoping to achieve/learn?
- Provide an account of existing approaches to the problem
- You can also preview any interesting results

The body (6~8 slides)

- Develop your main message (“... by using a variation of the minimax algorithm...”)
- Answer the “how” questions
 - How did you attempt to solve the problem?
 - Assume low level of familiarity with specific methods
 - Don’t expose too much detail
 - Focus on input/output, main components. Use diagrams if appropriate
 - How did you evaluate your results?
 - Explain your main evaluation scenarios
 - Show your results (tables, graphs...)
 - Show a demo if appropriate

The closing (1~3 slides)

- Restate your main message (“Our minimax tic-tac-toe agent shows/achieves/teaches us...”)
- Answer the “so what” question
 - How do your results compare with your expectations?
 - How do your results compare to existing approaches?
 - What did you learn? What surprised you?
 - What would you do differently? How would you continue developing it?
- Leave the audience with 1~3 ideas they should remember from your talk

Other hints

- Avoid text-heavy slides
 - Keywords rather than complete sentences (for the most part)
 - Leverage animations, graphics
 - No small / low contrast fonts
 - If a slide still ends up text-heavy, quickly state the main point
 - Try not to read directly from it

Other hints

- Use figures, diagrams, tables... to support your points
 - Elements (axes, columns) should have a descriptive label
 - Avoid cluttered visuals
 - If you can't avoid a small font (when copying from another source), try to highlight it somehow
 - Give credit if appropriate

Notes on absence

- Presentations should be delivered **in person** by the **whole group**
- Absence will only be allowed for excusable reasons according to university guidelines (sickness included)
- If you or a team member needs to be excused, write to me **as soon as you become aware of this**
- Plan for how this will impact your group: the team member should still be part of the presentation (either on a live call or pre-recorded).

Other hints

- Rehearse!
 - You don't want to be reading from your slides
 - Using them as memory aids is ok
 - You want to have a good handle on time (12-15 minutes)
 - For group talks, each team member will talk for around 3~5 minutes. It shouldn't take too long to rehearse your part individually
 - Still, make sure you also rehearse together!